

WASP-140b

R-2105

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-140_240117 · created 2026-08-10T23:45:51+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460326.6935 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1532 +/- 0.007
Transit depth (Rp/Rs)^2	2.35 +/- 0.22 [%]
Orbital Inclination (inc)	84.31 +/- 0.2
Airmass coefficient 1 (a1)	1.2539 +/- 0.006
Airmass coefficient 2 (a2)	-0.1144 +/- 0.0028
Scatter in the residuals of the lightcurve fit is	0.48 %
Best Comparison Star	None
Optimal Aperture	4.65
Optimal Annulus	10.86
Transit Duration (day)	0.0761 +/- 0.0026

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1532 $\pm$ 0.007 vs 0.1432 (deviation 1.43 σ)
Duration fit vs published	0.0761 d vs 0.0604 d (deviation 6.03 σ)
Mid-time O–C	-9.5 min
Depth SNR	21.9
Residual scatter	0.48 %

Light curve

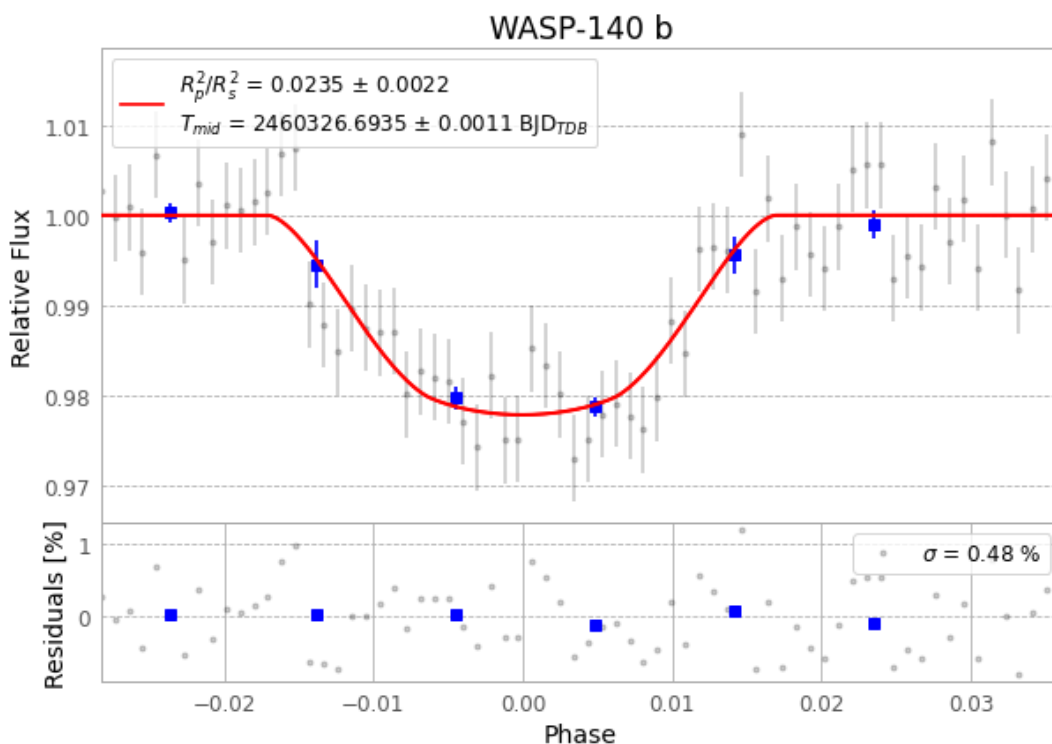


Figure 1 — FinalLightCurve_WASP-140 b_2024-01-16.png (SHA-256 ff3fb7cc474706f4...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [435, 236], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 77e4cc9795e0488d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

70 FITS frames (46 MB), WASP-140240117030316.FITS → WASP-140240117063015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-140-240117.log (7a86c551aae0...), FinalLightCurve_WASP-140 b_2024-01-16.png (ff3fb7cc4747...), FinalLightCurve_WASP-140 b_2024-01-16.csv (7d15315a28d0...)

Compute resources

Wall time 207.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-10 23:45Z.